

UPI Offline Mesh Network & Settlement System

Complete Hinglish Reference Guide & Complete Discussion Notes

1. System Kya Hai Aur Kaise Kaam Karta Hai? (From Zero)

Problem Statement: India me normal UPI payments ke liye active 4G/5G/Wi-Fi connection chahiye hota hai. Basement parking, underground metro, crowded stadiums, hilly areas, ya natural disaster me cellular network fail ho jata hai.

Solution: Offline Mesh System payments ko sender ke phone par offline sign aur encrypt karta hai. Phir Bluetooth Low Energy (BLE) Mesh Network ke zariye packet hop-by-hop pass hota hai jab tak kisi phone ko internet na mil jaye.

Step-by-Step Workflow:

Step 1: Offline Payment Create Karna: Alice ₹500 ka payment instruction banati hai. Application ek random 256-bit AES key generate karta hai, JSON payload ko AES-256-GCM se encrypt karta hai, aur AES key ko Bank ki RSA-2048 Public Key se encrypt karta hai (Hybrid Encryption). Intermediate phones sensitive data nahi dekh sakte.

Step 2: Mesh Propagation (BLE Hopping): Packet Bluetooth se nearby phones ko pass hota hai. Har hop par TTL (Time To Live) 1 se kam hota hai (e.g. 5 to 4). TTL 0 hone par packet stop ho jata hai.

Step 3: Bridge Node Connection: Koi ek stranger ya bridge phone jab internet zone (4G/Wi-Fi) me aata hai, toh background me packet POST /api/bridge/ingest par upload ho jata hai.

Step 4: Server Security & Settlement: Server SHA-256 hash nikal kar Redis SET NX EX atomic gate se duplicate drop check karta hai. RSA private key se AES key decrypt karke AES-GCM tag verification (tamper check) aur timestamp freshness check karta hai. Settlement me Alice ₹500 debit aur Bob ₹500 credit hota hai.

2. Real-Life Use Cases

- **Underground Parking & Metros:** Basement parking charges ya metro kiosk payment jahan network 0 hota hai.
- **Crowded Concerts & Stadiums:** 50,000 log ek jagah hone par jab mobile towers jam ho jate hain.
- **Remote Villages & Hilly Areas:** Mountain trekking ya rural areas jahan cellular connectivity nahi hoti.
- **Natural Disasters:** Earthquake ya Flood me jab cell towers crash ho jate hain.

3. Real Life Me Apply Hoga Ya Nahi? (Production Reality Check)

Code Level (YES): Hybrid RSA+AES Cryptography, Redis SET-NX Idempotency, Mesh TTL Gossip, aur Database Transactions 100% production-standard hain.

Real commercial banking launch ke liye 4 external requirements chahiye hong:

1. **NPCI & RBI Approval:** Offline wallet risk limits (e.g., ₹200-₹500 max per offline transaction).
2. **Hardware Security (TEE / Secure Enclave):** Keys phone ke ARM TrustZone / Hardware Chip me store hong.
3. **Mobile OS BLE Background Scanning:** iOS aur Android background BLE permissions handling.
4. **Offline Wallet Reserve:** Account se pehle se ₹2,000 reserve/lock rakhna (NPCI UPI Lite model).

4. Project Resume & Placement Value

Yeh project ordinary resume level ka nahi, balki **TOP 1% S-TIER (High-Impact) RESUME PROJECT** hai for Backend, Full-Stack, FinTech, and Systems Engineer roles.

UPI Offline Mesh – Cryptographic Settlement & BLE Mesh Simulator | MERN, Redis, Docker, Cryptography

- Designed an offline UPI payment settlement engine using RSA-2048-OAEP and AES-256-GCM hybrid encryption.
- Built a multi-hop Bluetooth Low Energy (BLE) virtual mesh simulator with TTL hop decay.
- Engineered a zero-race-condition idempotency layer using atomic Redis SET NX EX keys on SHA-256 digests.
- Developed an ACID-compliant MongoDB transactional settlement engine and Socket.IO real-time dashboard.
- Containerized monorepo with Docker Compose and achieved 100% pass rate across Jest test suites.

5. Interactive UI Dashboard User Guide (<http://localhost:5173>)

- **Inject Payment:** Alice (■5,000) → Bob (■1,000) ■500 ka packet banakar phone-alice par hold karta hai.
- **Run Gossip Round:** BLE packet nearby phones ko pass hota hai aur TTL decay hota hai.
- **Flush Bridge Node:** 4G bridge node packet backend ko bhejta hai. Alice ■4,500 aur Bob ■1,500 ho jata hai.
- **Tamper Ciphertext Demo:** Ciphertext ka 1 byte modify karke AES-GCM tag failure verify karta hai (INVALID).
- **Replay Attack Demo:** Identical packet dubara bhejne par Redis idempotency drop karta hai (DUPLICATE_DROPPED).
- **Simulate 3 Concurrent Bridges:** 3 simultaneous POST requests me exactly 1 SETTLED aur 2 DUPLICATE_DROPPED hote hain.

6. Can it be a Website for Real Offline Payments? (PWA Roadmap)

YES! Native App ke bina sirf Website (PWA) se bhi Real Offline Payments possible hain using modern Web APIs:

- **PWA & Service Workers:** Airplane Mode me bhi website browser me open hogi.
- **Web Crypto API (`window.crypto.subtle`):** Browser ke andar RSA-2048 + AES-256-GCM hardware encryption.
- **Web Bluetooth API (`navigator.bluetooth`):** Chrome browser paas ke Bluetooth devices se offline connect kar sakta hai.
- **Service Worker Background Sync:** Net aate hi background me Razorpay / Cashfree / Bank API Call execution.